$y_p' = 2Be^{2t}$ $\quad y_p'' = 4Be^{2t}$

and substitute into the nonhomogeneous problem:

$$4Be^{2t} - 3\cdot 2Be^{2t} - 4Be^{2t} = 3e^{2t}$$

$$\Rightarrow B = -1/2$$

$$y_p = -\frac{1}{2}e^{2t}$$

The general soln is

$$y = y_c + y_p = c_1e^{-t} + c_2e^{4t} - \frac{1}{2}e^{2t} \quad //$$

Example 3.23 $\quad y'' + 4y' + 4y = t^{-2}e^{-2t}$

1/ homogeneous problem: characteristic eq^n $m^2 + 4m + 4 = 0$

$\Rightarrow$ repeated roots $m_1 = m_2 = -2$

Reduction of order gives the fundamental set of soln^s:

$$y_1 = te^{-2t} \qquad y_2 = e^{-2t}$$

2/ $g = t^{-2}e^{-2t}$ $\qquad W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} te^{-2t} & e^{-2t} \\ te^{-2t}(-2) + e^{-2t} & -2e^{-2t} \end{vmatrix}$

$$= -e^{-4t} \quad (\neq 0)$$

3/ $u_1 = -\int_{t_0}^{t} \frac{y_2 g}{W}\, ds$

$$= -\int_{t_0}^{t} \frac{e^{-2s}\, s^{-2}\, e^{-2s}}{-e^{-4s}}\, ds$$

$$= \int_{t_0}^{t} s^{-2}\, ds$$

$$= -\left[s^{-1}\right]_{t_0}^{t} = t_0^{-1} - t^{-1} \quad //$$

$$u_2 = \int_{t_0}^{t} \frac{y_1 g}{W}\, ds$$

$$= \int_{t_0}^{t} \frac{te^{-2t}\, t^{-2}e^{-2t}}{-e^{-4t}}\, ds \qquad \int_{s=t_0}^{t} -s^{-1}\, ds$$

$$= -\left[\ln|s|\right]_{t_0}^{t} = \ln|t_0| - \ln|t| = \ln t_0 - \ln t \text{ since } t > 0$$

2/10/26

Nonhomogeneous Second Order ODEs : Review

$$y'' + p(t)y' + q(t)y = g(t)$$

1/ Solve homogeneous eqⁿ to find complementary solⁿ $y_c = c_1 y_1(t) + c_2 y_2$
2/ Find a particular solution of nonhomogeneous problem
- How?

- MuC
- VoP

Ex 3.23 cont'd

$$y_p = -y_1 u_1 + y_2 u_2$$

$$= -te^{-2t}\left(\ln t_0^{-1} - t^{-1}\right) + e^{-2t}\left(\ln t_0 - \ln t\right)$$

$$= +e^{-2t} - e^{-2t}\ln t - t_0^{-1} t e^{-2t} + \ln t_0 e^{-2t}$$

and the general solution is

$$y = y_p + y_c$$

$$= -e^{-2t} - e^{-2t}\ln t - t_0^{-1} t e^{-2t} + \ln t_0 e^{-2t} + c_1 t e^{-2t} + c_2 e^{-2t}$$

Reduction of Order

Example 3.24

$$y'' - \frac{(4t+1)}{t}y' + \frac{2}{t}(2t+1)y = 4e^{2t}$$

- Check $y_1 = e^{2t}$ is a solution of the homogeneous problem

$$y_1' = 2e^{2t}$$
$$y_1'' = 4e^{2t}$$

$$\text{LHS ODE} = 4e^{2t} - \frac{(4t+1)}{t}2e^{2t} + \frac{2}{t}(2t+1)e^{2t}$$

$$= e^{2t}\left[4 - 8 - \frac{2}{t} + 4 + \frac{2}{t}\right]$$

$$= 0 \quad \text{ie } y_1 \text{ is a sol}^{\text{n}} \text{ of the homogeneous problem.}$$

Set $y = v(t)\, y_1(t)$
$= v(t)\, e^{2t}$

$y' = v e^{2t} 2 + e^{2t} v'$ (product rule)

$y'' = 2v e^{2t} \cdot 2 + e^{2t} 2v' + e^{2t} v'' + v' e^{2t} \cdot 2$

$= e^{2t} [v'' + v' 4 + 4v]$

Now substitute into the inhomogeneous eq^n:

$e^{2t}[v'' + 4v' + 4v] - \frac{(4t+1)}{t}\{2v + v'\} e^{2t} + \frac{2}{t}(2t+1) v e^{2t} = 4e^{2t}$

$v'' + v'\left\{4 - \frac{(4t+1)}{t}\right\} + v\left(4 - \frac{2}{t}(4t+1) + \frac{2}{t}(2t+1)\right) = 4$

$\Rightarrow v'' + v'\left\{4 - 4 - \frac{1}{t}\right\} + v\left(4 - 8 - \frac{2}{t} + 4 + \frac{2}{t}\right) = 4$

$\Rightarrow v'' - \frac{1}{t} v' + v(0) = 4$

Set $w = v'$ we have $w' - \frac{1}{t} w = 4$ ie (3.40)

a first order eq^n for $w(t)$

$w' + p(t) w = g$ with $p = -\frac{1}{t}$ $g = 4$

Integrating factor: $\mu = e^{\int p\, dt} = e^{\int -\frac{1}{t} dt} = e^{-\ln t} = e^{\ln t^{-1}} = t^{-1}$

Multiply eq^n by μ:

$\mu w' - \frac{1}{t} w \mu = 4\mu$

$\Rightarrow t^{-1} w' - t^{-2} w = 4t^{-1}$ since $\mu = t^{-1}$

$\Rightarrow \frac{d}{dt}(t^{-1} w) = 4t^{-1}$

Integrate: $t^{-1} w = \int 4t^{-1}\, dt = 4 \ln t + c$ $t \in (0, \infty)$

$\Rightarrow w = 4t \ln t + ct$

$\Rightarrow \frac{dv}{dt} = 4t \ln t + ct$ since $w = v'$

Integrate:

$v = 4\left\{\int t \ln t\, dt + \int c t\right\} dt + d$ (d constant)

Now $\int \overbrace{t}^{dv}\overbrace{\ln t}^{u}\,dt = \ln t \cdot \frac{t^2}{2} - \int \frac{t^2}{2}\frac{1}{t}\,dt$ integrate by parts

$$= \frac{t^2}{2}\ln t - \int \frac{t}{2}\,dt$$

$$= \frac{t^2}{2}\ln t - \frac{t^2}{4}$$

hence $v = 4\left(\frac{t^2}{2}\ln t - \frac{t^2}{4}\right) + \frac{ct^2}{2} + d$

$= 2t^2\ln t + c_1 t^2 + c_2$ for constants c_1 and $c_2 = d$

$c_1 = \frac{c}{2} - 1$

$\therefore \quad y = v(t)y_1$

$= \left(2t^2\ln t + c_1 t^2 + c_2\right)e^{2t}$ //

G.S. $y = c_1 y_1 + c_2 y_2 + y_p$ i.e. $y_1 = e^{2t}$

$y_2 = t^2 e^{2t}$

$y_p = 2t^2 \ln t\, e^{2t}$